

One stop shop. Turnkey manufacturing services are the need of the hour to fulfil the increasing demand for one-stop shops for EMS. However, turnkey projects need a high degree of technical skills that can be customised according to specific requirements of the OEMs.

Only a few Indian EMS providers offer turnkey solutions for PCBAs, utilising chip-on-board (COB), surface-mount (SMT) and through-hole technologies. They also provide complete assemblies, including plastic mouldings, metal-die castings and sheet metal fabrications, apart from finishing, painting and printing—in short, delivering the final assembly of completed units. This opens up vast business expansion opportunities for Indian EMS companies.

Original design manufacturing. Electronics products need constant design revisions as end-users expect creativity and continuous innovation. Therefore consumer electronics product design and development is often outsourced to original design manufacturers (ODMs). In such cases, the sooner an OEM engages the contract manufacturer for product design and development services, the better—particularly when the product being designed moves into production and ramp-to-volume phases.

Moreover, ODMs can ensure unique products as the Indian market needs combining of Design in India with Make in India. Hence, ODM services become an important and cost-effective solution for the customers.

According to a report published by PricewaterhouseCoopers (PwC), “Many EMS companies are expanding into new services that offer high margins. They are providing new services along the spectrum. At one end they are offering more design services, for either sub-assemblies or finished products, while at the other end they are offering more testing services. The more services EMS providers offer, the more they move into new models of joint design manufacturing and outsourced design manufacturing.”

Reverse logistics. Considering the importance of electronics waste management as well as after-sales services, including repair and main-

EMS company categorisation

By revenue

- Tier 1: >US\$ 5 billion
- Tier 2: US\$ 500 million to US\$ 5 billion
- Tier 3: US\$ 100 million to US\$ 500 million
- Tier 4: <US\$ 100 million

By operational models

- **High-volume, low-mix (HVLM).** A contract manufacturing setup where only a few assemblies are produced in large quantities is a production arrangement that may last for weeks or even months using the same setup. This technique allows change-overs to be kept at minimum levels and equipment utilisation rates to be significantly high. Contract manufacturers have proven to be more efficient when running at high volumes, which require minimum engineering changes. HVLM model is applicable in computers and peripherals, consumer electronics devices, telecommunication and storage devices.
- **High-mix, low-volume (HMLV).** This type of contract manufacturing puts a high focus on quality and customisation as per customer requirements. Considering high margins and niche market scenarios, even major changes in market dynamics often do not heavily impact such a production process. OEMs that prefer such solutions are willing to pay a higher price without compromising on quality. Recent industry trends show a shift towards this model of production, but success of this model depends on controlling and improving supply chain efficiencies. HMLV is applicable for industrial, medical, and aerospace and defence (A&D).

*Mix refers generally to the complexity or different models of the PCB assembly. Volume refers to the number of units built, with products like consumer electronics on the high end; while prototype, medical electronics or machinery on the low end.

Very few companies operate with both HMLV and HVLM models, simultaneously.

tenance, reverse logistics has become an important activity.

Global OEMs also prefer outsourcing reverse logistics. Services related to repair/rework and refurbishment work will not only help EMS firms get additional business from OEMs but also enable them to play a role in e-waste management.

Going green with processes and products. There is a pressure on manufacturers to adopt green manufacturing processes while simultaneously producing electronics that are less expensive to use thanks to lower energy demands. Environmental concerns are paramount not only during the manufacturing process but throughout the lifespan of electronic devices. This, in turn, is compelling manufacturers to develop new methods to produce devices that consume less energy.

Moving forward

The ever-increasing end-users' demand, and fast-paced technological developments, compel OEMs to continuously introduce new and innovative products in the electronics market. Consequently, OEMs have to increasingly depend on EMS providers who offer significant benefits such as cost savings, reduced time-to-market, reduced time-to-volume,

quality and flexibility.

EMS companies operate as strategic partners of OEMs by providing them with a full range of services, including contract design, prototyping, final system assembly, configuration, order fulfilment and even after-market services, including repair. By using the services of EMS providers, OEMs can concentrate on their core competencies such as research and product development, brand building, sales and marketing.

Outsourcing to EMS providers also enables OEMs to gain access to the latest equipment, process knowledge and manufacturing know-how without having to make substantial capital investments, as risks are converted into variable costs. A strategic partnership between the two can enable EMS providers as well as OEMs to contribute effectively to the success of the end-product. This, in turn, helps both parties achieve high profitability and a good market share.

Since OEMs are quite selective in choosing their EMS partners, the latter needs to focus on nurturing long-term relationships with their customers through enhanced value-added services, strategic partnerships and alliances, as well as diversification. **EFY**